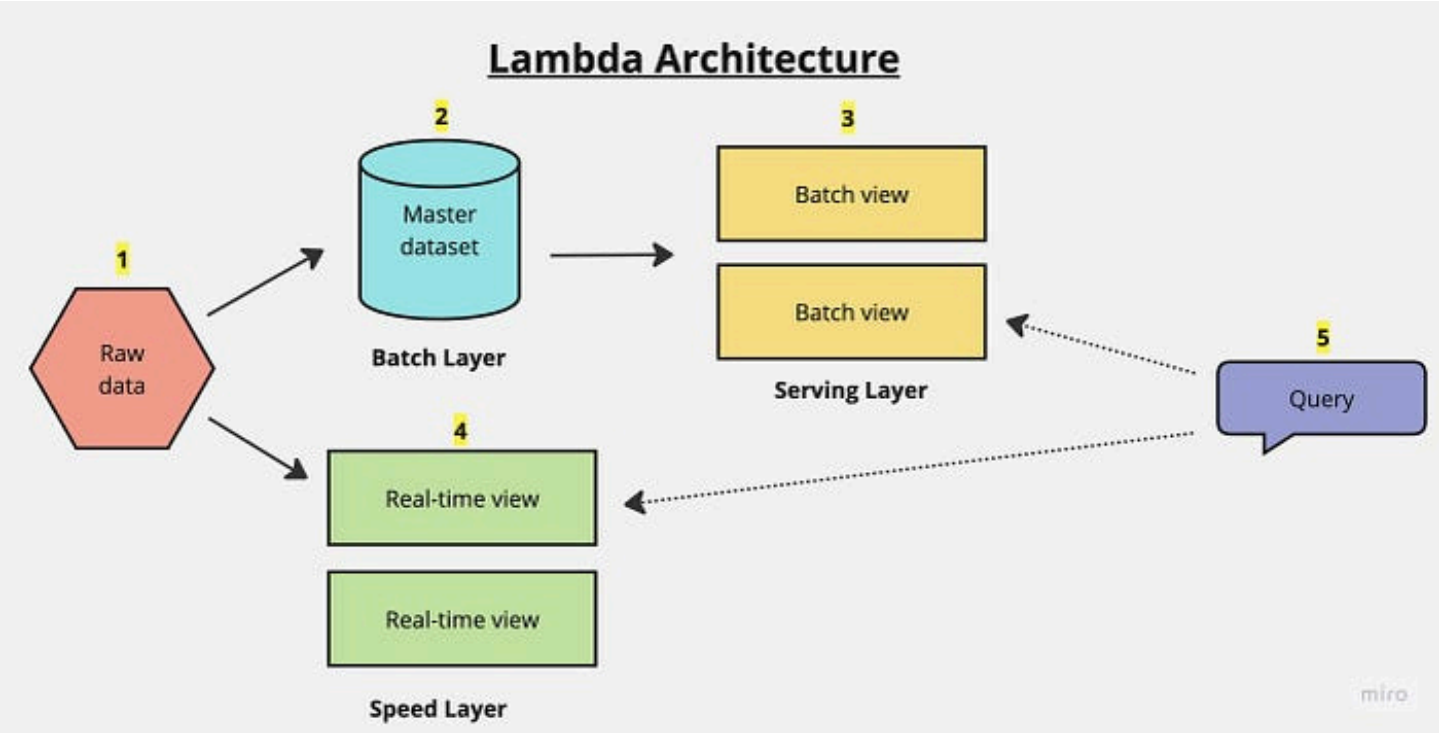
Venturing into the world of designing and building robust systems for big data collection or analysis can be daunting. For those who are curious about tech, I have listed some of the main data architecture structures in jargon-friendly descriptions.

Lambda Architecture

Lambda architecture is a data-processing architecture comprised of three-layers — a batch layer (data processing at rest), a speed (or real-time; data processing in motion) processing layer, and a serving layer (to respond to queries). Systems here operate **independently** of each other. Colossal amounts of real-time data is managed by batch processing and stream processing procedures, involving a traditional batch data pipeline and a rapid streaming data pipeline.

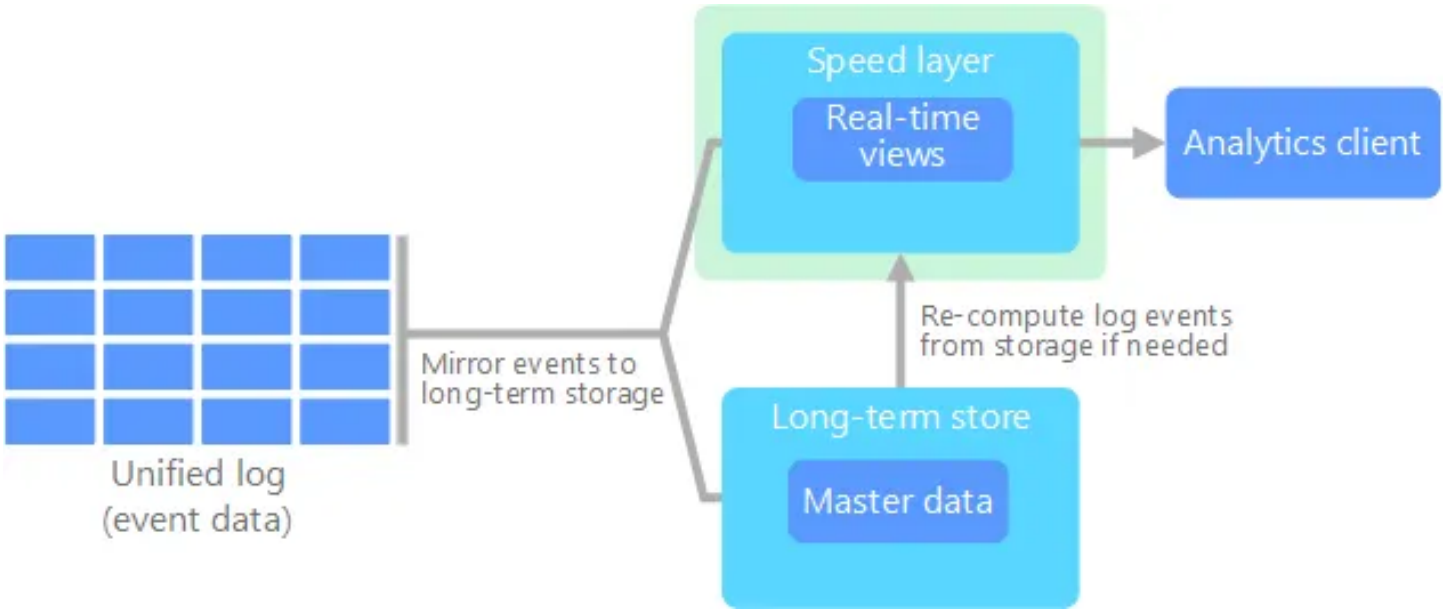
Initially in the batch layer, datasets are stored in terabytes or petabytes as immutable but append-only status, and precomputed. This allows records of constant incoming data to be structurally reliable. Second, the serving layer receives data from the batch layer on a set schedule and also real-time data streaming in from the speed layer. As a result, the serving layer queues incoming data for the next scheduled run. Lastly, the speed processing layer mitigates the gap between time of data creation and when the data is available for querying.



One main disadvantage of using the Lambda architecture is managing multi-layer independent systems which increases the chances of systemic error. Here is where Kappa architecture arrives at the scene.

Kappa Architecture

Kappa architecture is true event-base, so it does real-time streaming and batch processing in a single path using a stream processing system. It differs from the Lambda architecture in a sense that it avoids maintaining two different codebases between the batch and speed layers. The key purpose is using a single stream processing engine instead of Lambda's multi-layered architecture to process real-time data. The events are ingested from a stream, analyzed, and created as new events in new streams. Kappa was created for better streamlining by Jay Kreps to address the shortcomings of the Lambda architecture, but it does not replace Lambda due to being costly and being difficult for implementing data replay.



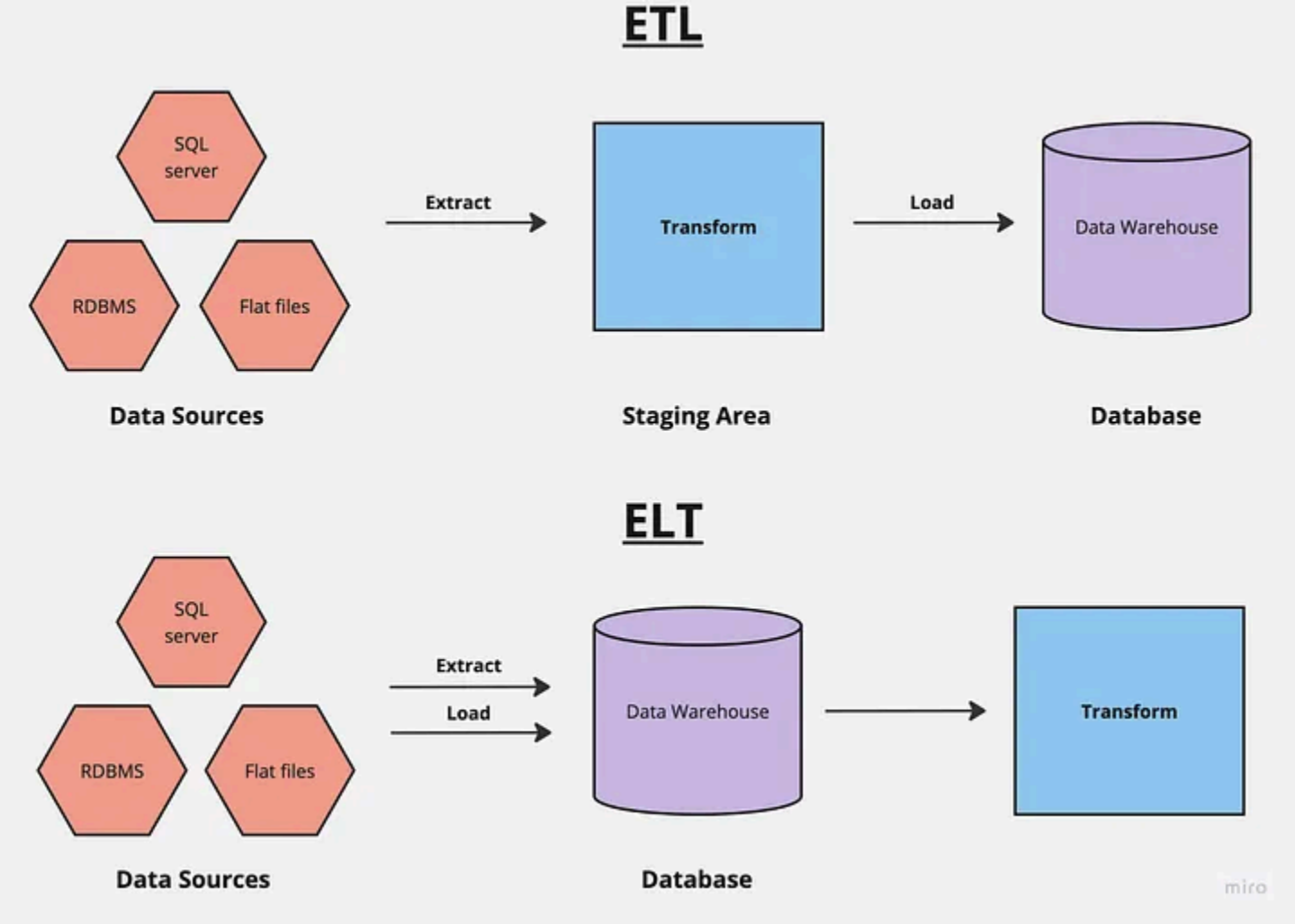
Data Warehouse

A data warehouse is one of the oldest and most well-established data architectures. It is highly formatted and structured for use in analytics. Before data sources are stored in the warehouse, they must undergo either an **ETL** or **ELT** pipeline.

An **ETL** pipeline involves extracting, transforming, and loading data. Raw acquired data can be derived from databases such as the SQL server,

relational database management system (RDBMS), or flat files. For this instance, Apache Airflow would be an excellent open-source workflow tool.

A newer variation of the **ETL** pipeline is the **ELT**. An **ELT** pipeline involves extracting, loading and then transforming. Essentially, raw acquired data is loaded **directly** into a database (like a data warehouse) before it is transformed.



Data Mesh Architecture

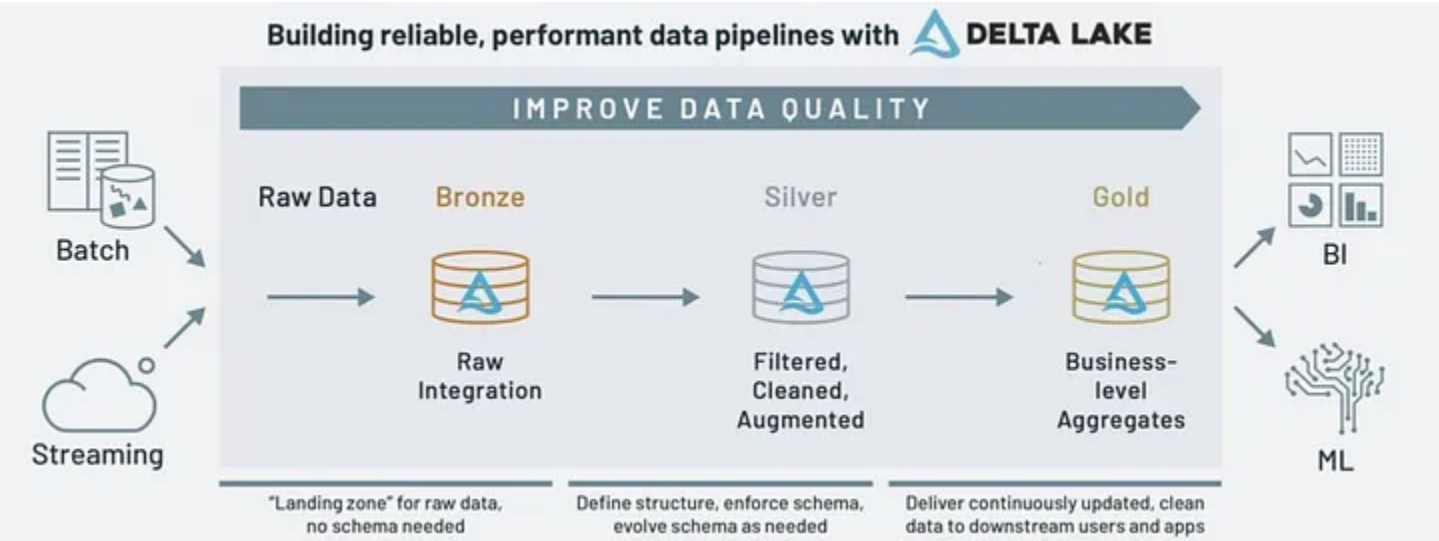
Data mesh architecture is a concept that solves advanced data security challenges via distributed, decentralized ownership. By breaking up data pieces, it empowers business units to have autonomy and ownership over their data domain because a data platform architecture becomes inundated over time. A data mesh architecture is advantageous where it has increased flexibility to modify, is less costly (real-time data streaming is cheaper than batch processing), and improves data security.

Data Lake

Data lake architecture is like a data warehouse except that it contains **raw formatted** and **unstructured data**. This is analogous to brainstorming and dumping all your ideas onto a visionary dashboard.

Data Lake + Data Warehouse = **Data Lakehouse!**

A data lakehouse is the next generation architecture as a result of the converging the data lake and data warehouse. This open system uses similar data warehouse managing features in conjunction with low-cost storage used for data lakes. A couple of big dog data lakehouse companies are Snowflake and Databricks.

Medallion Architecture

The medallion architecture is implemented to logically organize data in a lakehouse. It is provided by Databricks in three layers of data quality — bronze, silver, and gold.